

KABITA DHAR

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Summary

Dedicated AI/ML Engineer bridging the gap between advanced machine learning research (GNNs, RL) and production-ready software. Leveraging a strong foundation in C++ and algorithmic optimization to architect scalable, low-latency intelligent systems. Proven experience in deploying AI-driven threat detection models and optimizing computational networks.

Education

North Eastern Regional Institute of Science and Technology (NERIST)	August 2023 – June 2026
• BTech in Computer Science and Engineering	
PCPS Girls' Polytechnic, Guwahati	July 2020 – June 2023
• Diploma in Computer Science and Engineering	
Coursework: Data Structures, Algorithms, Data Science, Software Engineering, Machine Learning,	

Projects

Federated Attention-Based Graph Neural Network for Malicious User Detection	github.com
• Developed a federated attention-based graph neural network framework that detects malicious users in dynamic cognitive radio networks by collaboratively learning spatial and temporal network behavior while preserving data privacy.	
Taxi Route Optimization with Reinforcement Learning	github.com
• Built a Q-learning-based Reinforcement Learning agent for the Taxi-v3 environment, using an -greedy policy and Bellman updates to efficiently optimize navigation, passenger pickup, and drop-off.	
Real-Time Network Threat Detection System	github.com
• Built a real-time network threat detection system that trains an anomaly-detection model in Python, exports it to ONNX, and runs low-latency inference from a multithreaded C++ pipeline. The project combines machine learning, feature engineering, and systems design to detect suspicious network behavior in real time.	
Sudoku Solver	github.com
• Developed a Sudoku Solver application using a backtracking algorithm in C++ to efficiently solve 9x9 puzzles with real-time user input support.	

Internship Experience

CyberSecurity Intern, MNIT - Allahabad	May 2025 - June 2025
• Developed and implemented a Hybrid AI Against Ransomware using CNN, MCTS, and GNN algorithms, identifying threats with a 99% accuracy rate on real-time threat datasets. github.com	
Full Stack Development Intern, IIT - Guwahati	June 2023 – Aug 2023
• Engineered the full-stack architecture for the official IIT Guwahati website serving 5,000+ students and faculty, reducing content update time by 40% through the implementation of dynamic front-end components and optimized Node.js/Python backend workflows. github.com	

Technical Skills

Programming & Core Concepts:	C++, Python, SQL, Data Structures & Algorithms
Machine Learning & Deep Learning:	Large Language Models (LLMs), Transformers, RAG, GNN, CNN, RNN, Reinforcement Learning
AI Domains:	Generative AI, Natural Language Processing (NLP), Computer Vision, Cybersecurity ML, Network Intelligence, Time Series Modeling
MLOps, Cloud & Tools:	Docker, FastAPI, Git/GitHub, CI/CD Pipelines, Oracle Cloud Infrastructure (OCI), AWS, Model Deployment, Performance Tuning

Courses and Certifications

Software Engineering Job Simulation, JPMorgan Chase & Co.	October 2025
Cloud Infrastructure Generative AI Professional, Oracle	October 2025
Amazon ML Summer School , Amazon	May 2024
Ethical Hacking WorkShop, IIT - Bombay	June 2021

Achievements

Training and Placement Secretary, NERIST - Nirjuli	Present
National Cadet Corps, NERIST - Nirjuli	Present
Head of IT Team, Model United Nation	May 2024
Annual College Hackathon (HackShristi - NERIST), Secured 1st among 30 teams	April 2024
Smart India Hackathon (SIH), Participated at the All-India Level.	September 2024